

Cian Cullinan Systems and software engineer with many years of experience in designing, running, and optimising extremely large distributed systems.	+353 87 7997654 cian.cullinan@gmail.com www.linkedin.com/in/cian-cullinan
EXPERIENCE Google Ireland Staff Systems Engineer (SRE) Ranging across high impact projects spanning multiple teams and directing the work of other engineers. Dec 2022 - PRESENT • Rewrote the carbon efficiency/intensity pipelines used by Google's Carbon Aware Computing to be simpler and more reliable using Apache Beam. Also introduced Pandera type checking to the wider codebase to prevent frequent dataframe schema mismatches. • In response to several impactful outages, drove work to align the needs of high priority ML jobs with the capabilities of Google's storage stack. • Led the successful Colossus SRE efforts for Google's DMA 5(2) commitments. • Led the successful deployment of Colossus as part of Google's new sovereign cloud. • Designed, implemented and led the migration of Google's archival storage service from an arbitrary metadata sharding to per-cluster sharding for resource savings, better reliability and easier provisioning across hundreds of clusters. SRE Manager SRE Manager for two of Google's largest storage services. Jan 2020 - Nov 2022 • Managed Colossus and a separate internal archival storage team. Had 15 reports at peak. • Switched back to the individual contributor ladder to focus on a set of high-priority multi-year projects (see above). Staff Systems Engineer (SRE), Tech Lead Sep 2018 - Dec 2019 Lead SRE for Colossus, Google's cluster-level filesystem. Brought in as a senior engineer to bolster and grow a junior team. • Worked on several large-scale projects to standardise storage infrastructure across multiple teams. Oct 2015 - Aug 2018 Lead SRE for Spanner, Google's globally distributed database. Led the service through a doubling in size and to general availability, followed by public launch of the cloud service. • Led the service through a doubling in size and to general availability, followed by public launch of the cloud service.	Education Trinity College Dublin, Ireland MSc. High Performance Computing 2000 - 2001 Trinity College Dublin, Ireland BA Mod, Experimental Physics 1996-2000 Skills Technical areas of expertise: distributed systems (especially storage), designing for the tradeoffs of reliability/performance/scalability /cost, monitoring, SLA definition and defence, resource planning, automation, outage root-cause analysis, data analysis. Google equivalents of: Kubernetes, Hadoop/Spark, NoSQL DBs, Cassandra, CockroachDB, Prometheus, Grafana Non-technical expertise: coordinating and leading multi-team projects, building and managing teams of engineers for critical services, mentoring and career guidance for SRE and Google scholars, interviewing, technical training, team planning and prioritisation, balancing operational and project work. Programming Languages Intermediate: Python, Go Basic: Shell, C, SQL

Systems Engineer (SRE)

Nov 2012 - Sep 2015

[Spanner](#) SRE

- Led multi-year project to encrypt all data at rest. Completed ahead of schedule with zero outages.

Jan 2011 - Oct 2012

[Google Analytics](#) SRE

Feb 2007 - Dec 2010

Core SRE - responsible for services necessary for running and securing aspects of Google data centres.

Openwindow — Full Stack Engineer

Oct 2004 - Jan 2007

A startup building video streaming systems for healthcare settings. Designed and built a Video on Demand service.

Servecast — System Administrator

Mar 2004 - Sep 2004

IPTV company, since acquired by Level 3. Linux administrator for streaming and archiving services.

Media Lab Europe — Research Engineer

Sep 2001 - Mar 2004

[Researching](#) and prototyping video conferencing, remote haptics and gesture based interfaces.